AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 2 · WORKBOOK

Features, Labels, Targets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One connected feature/target contract record.

Explain inputs, trace a target boundary, repair leakage and state a missing-snapshot policy.

Keep one habit: Keep feature order, units, timestamp/provenance, target rule and new-record uncertainty together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

- Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

MAIN LESSON · SECTION 12

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Library task at checkout: ordered features [pages,queue_at_checkout], positive integer pages and nonnegative queue; returned_late=1 iff completed days_to_return>14, exactly 14→0. Worked past 120/2/days 18→[120,2], label 1; new 80/0→[80,0], label unknown. W2 96/4 at 14 days→[96,4],0; at 15 days same features, target 1. Exclude later return/due-derived fee and target; arbitrary loan_id metadata.

  Plant task uses today_light/today_soil_moisture, not later observed tomorrow/next-week fields.

- Reference for this exercise: open this week's student lesson, "Features, Labels, Targets", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked loan table, identify the two candidate **feature**s, the **target label** and the tracking field. Explain why days_to_return is not a predictor.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Library task at checkout: ordered features [pages,queue_at_checkout], positive integer pages and nonnegative queue; returned_late=1 iff completed days_to_return>14, exactly 14→0. Worked past 120/2/days 18→[120,2], label 1; new 80/0→[80,0], label unknown. W2 96/4 at 14 days→[96,4],0; at 15 days same features, target 1. Exclude later return/due-derived fee and target; arbitrary loan_id metadata.

  Plant task uses today_light/today_soil_moisture, not later observed tomorrow/next-week fields.

- A completed loan has pages 96, queue_at_checkout 4 and days_to_return 14. Give the ordered **feature** list and **target**. Then state what changes if it took 15 days instead.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Library task at checkout: ordered features [pages,queue_at_checkout], positive integer pages and nonnegative queue; returned_late=1 iff completed days_to_return>14, exactly 14→0. Worked past 120/2/days 18→[120,2], label 1; new 80/0→[80,0], label unknown. W2 96/4 at 14 days→[96,4],0; at 15 days same features, target 1. Exclude later return/due-derived fee and target; arbitrary loan_id metadata.

  Plant task uses today_light/today_soil_moisture, not later observed tomorrow/next-week fields.

- A model has every test **prediction** correct after using pages and days_to_return. A learner says a separate test set makes this valid. Explain the flaw and repair the **input**s.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Library task at checkout: ordered features [pages,queue_at_checkout], positive integer pages and nonnegative queue; returned_late=1 iff completed days_to_return>14, exactly 14→0. Worked past 120/2/days 18→[120,2], label 1; new 80/0→[80,0], label unknown. W2 96/4 at 14 days→[96,4],0; at 15 days same features, target 1. Exclude later return/due-derived fee and target; arbitrary loan_id metadata.

  Plant task uses today_light/today_soil_moisture, not later observed tomorrow/next-week fields.

- For the robot task, a dataset stores queue_now at export time but has no saved queue_at_dispatch. Can you rename that field to make it usable? State a defensible next step.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the dispatch-time task's own field names, order and strict more-than boundary. Keep observed completed-trip labels separate from a new trip's unknown outcome. No future arrival measurement exists for the new trip, and no model prediction is supplied.

- A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

  Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

  For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Audit a new task independently

> **My independent explanation, trace and limits**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Define the Plant Classifier **target** and a first **feature** list; record units, allowed values, collection method, and availability time.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Trace the **feature contract** for **Feature**s **label**s and **prediction time**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Feature**s **label**s and **prediction time**. Proposed explanation: "Every stored column is a **feature**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Rows contain leaf length, edge type and healthy/spotted **label**.
- Identify **feature**s, **target** and **prediction**-time **input**s.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A row contains leaf_length_cm=8, leaf_width_cm=3 and species=fern. For predicting species, identify two **feature**s and the **target**.

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 2 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A table has leaf length, edge type and verified species. A model predicts species using all three columns. Identify leakage and a legitimate **input** set.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A fictional robot predicts at dispatch whether a parcel trip will take more than 10 minutes.

  Fields are route_steps and queue_at_dispatch captured at dispatch, trip_id for tracking, arrival_minutes measured after arrival, and slow = 1 when arrival_minutes > 10. Specify **input**s, order, **target** and exclusions.

  For a past trip with steps 30, queue 2, arrival 10, give **feature**s and **label**. For a new trip with steps 24 and queue 0, give **feature**s and explain whether slow is known.

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Choose **feature**s that exist when a plant **prediction** must be made.
- Predict next-week wilting at today's photo capture. Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/2

Features, Labels, Targets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Mark prediction time

The decision occurs today, before next week's outcome.

2. Select inputs

Today's light and soil readings are available features; plant_id identifies the plant.

3. Keep target separate

next_week_wilt supplies the training label, not an input available today.

CHANGE ONE THING

A table adds tomorrow_soil_moisture as an input.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict, check and repair**

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Choose **feature**s that exist when a plant **prediction** must be made.
- Which **label**led step matches this detail? next_week_wilt supplies the **training label**, not an **input** available today.
- 1. Select **input**s
- 2. Mark **prediction time**
- 3. Keep **target** separate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Keep **target** separate next_week_wilt supplies the **training label**, not an **input** available today.

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A table adds tomorrow_soil_moisture as an **input**. Which conclusion is supported by the supplied facts?
- 1. Reject it for this **prediction**-time contract.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Reject it for this **prediction**-time contract. Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Repair and keep the record

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, **label** your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

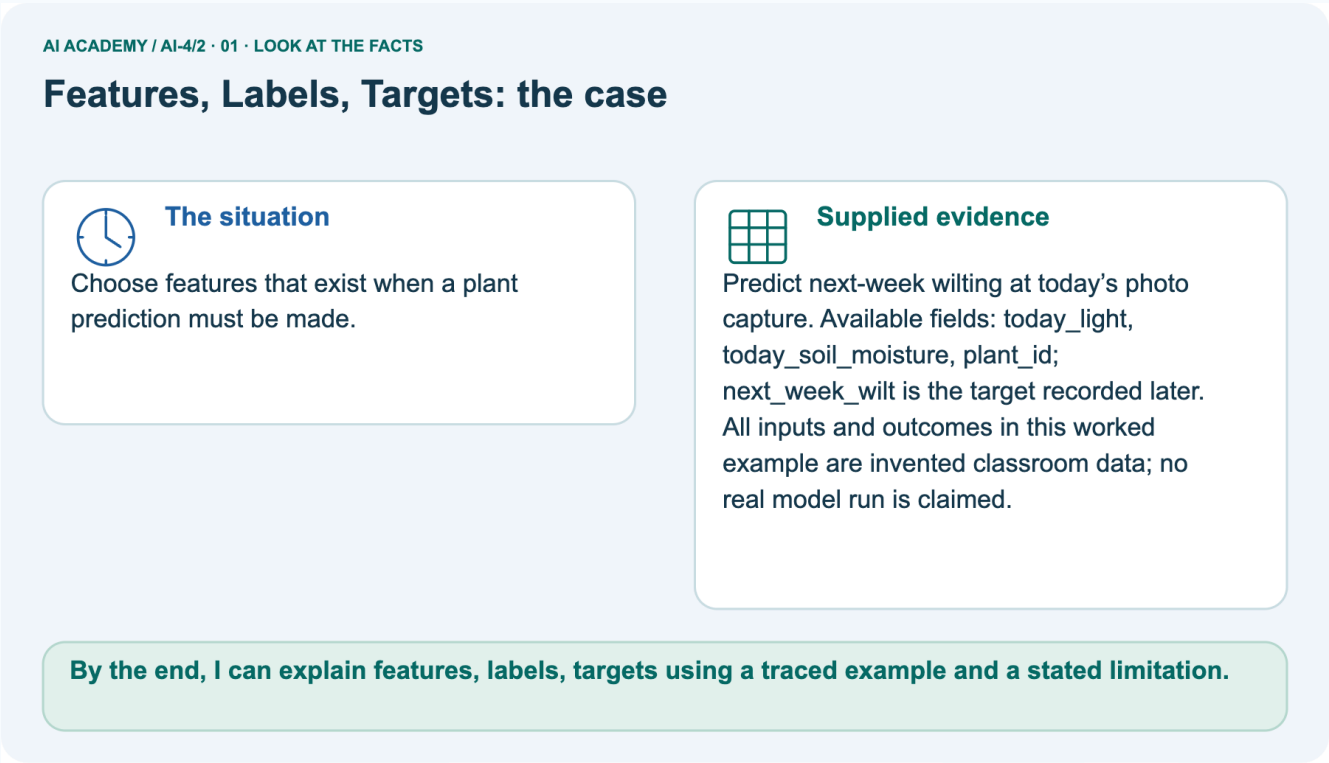
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Features, Labels, Targets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

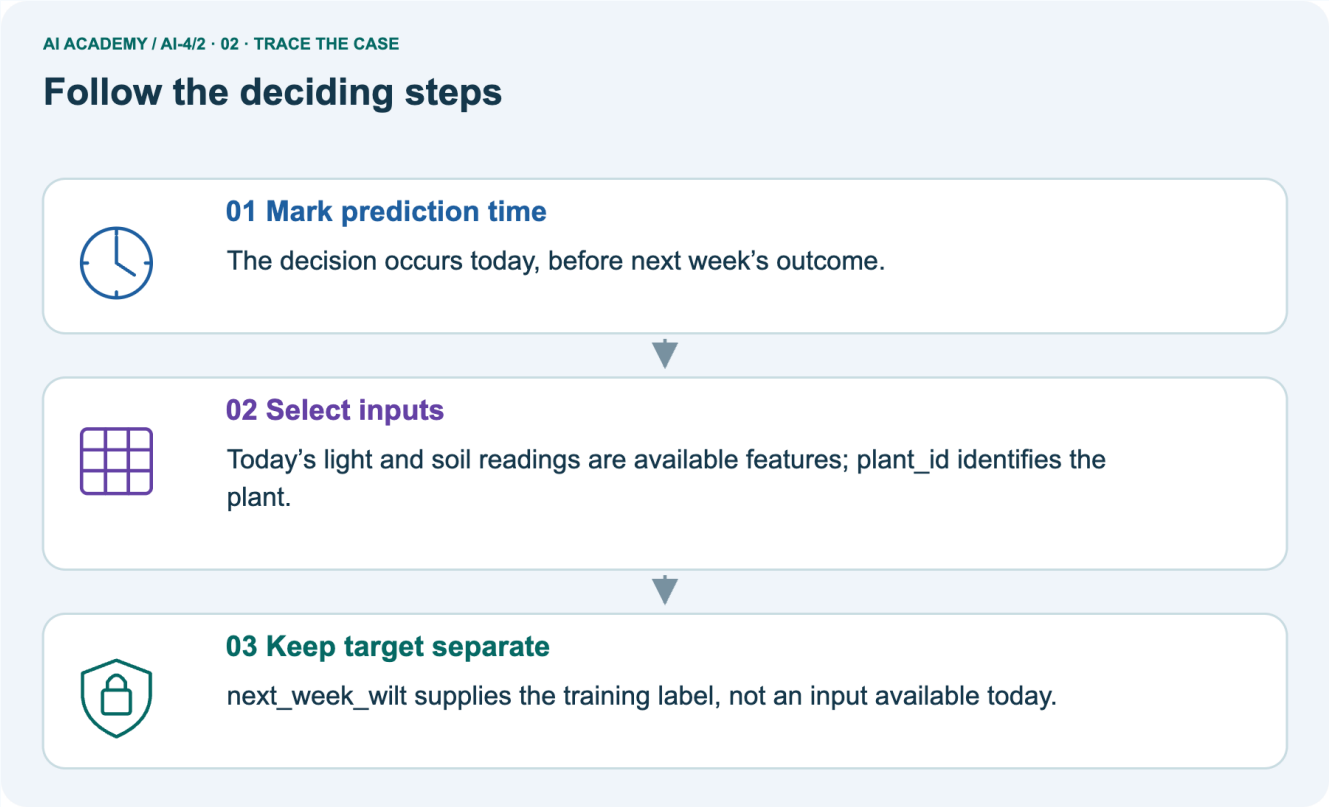
Choose **feature**s that exist when a plant **prediction** must be made. Supplied facts: Predict next-week wilting at today's photo capture.

Available fields: today_light, today_soil_moisture, plant_id; next_week_wilt is the **target** recorded later. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



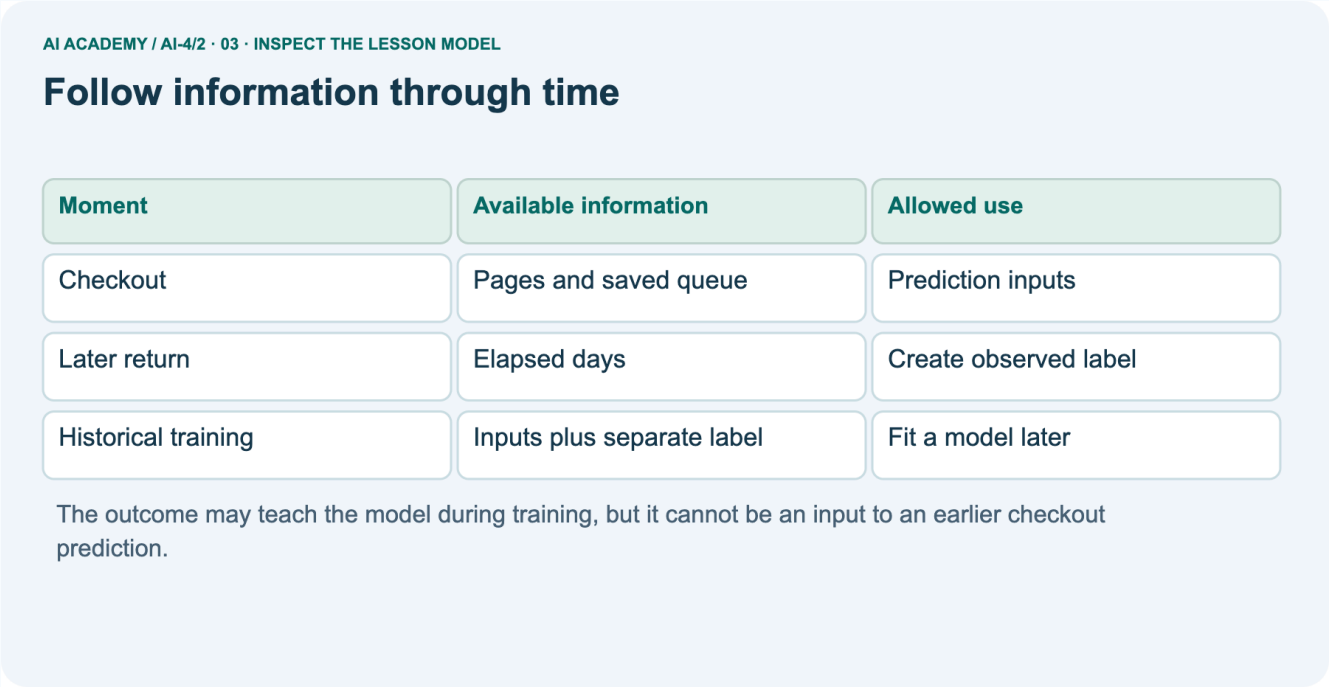
Follow the labelled picture with the matching explanation. Open the vector diagram

Mark **prediction time**: The decision occurs today, before next week's outcome. → Select **input**s: Today's light and soil readings are available **feature**s; plant_id identifies the plant. → Keep **target** separate: next_week_wilt supplies the **training label**, not an **input** available today.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow information through time



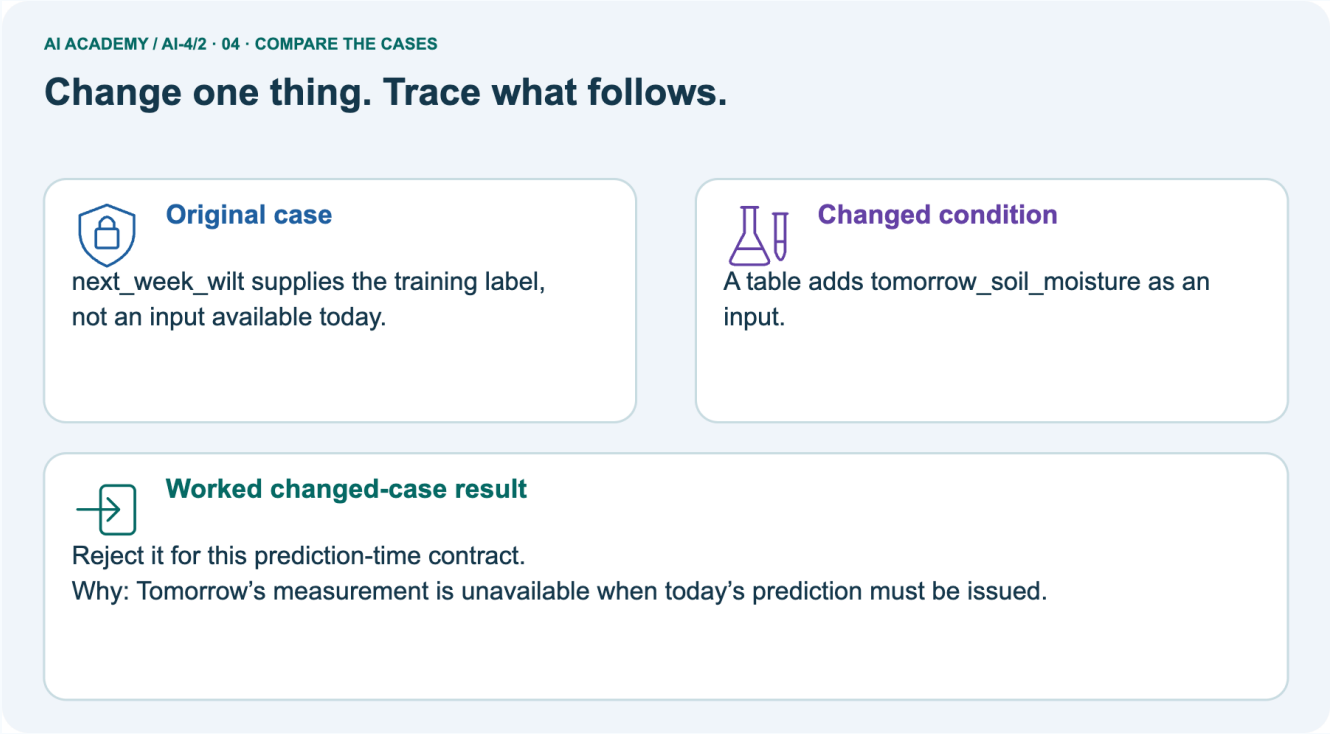
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow information through time. Checkout | Pages and saved queue | **Prediction input**s. Later return | Elapsed days | Create observed **label**. Historical **training** | **Input**s plus separate **label** | Fit a model later

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

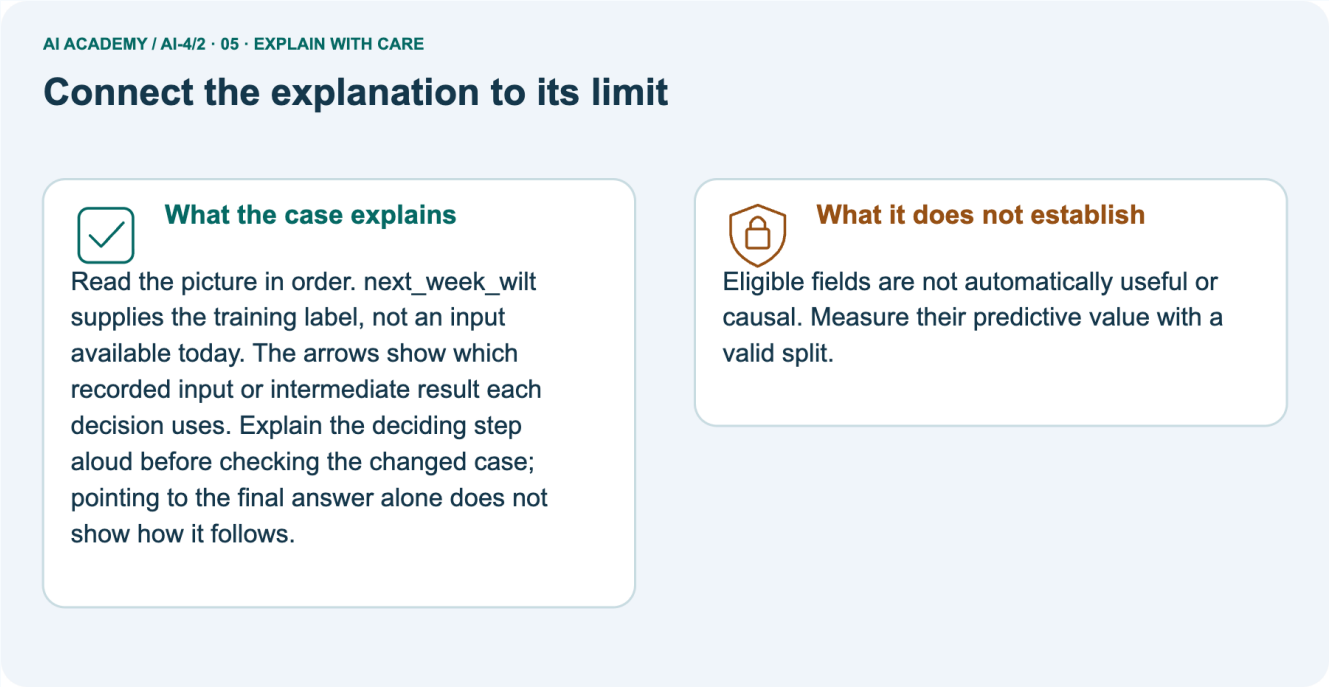
Original case: next_week_wilt supplies the **training label**, not an **input** available today. Changed condition: A table adds tomorrow_soil_moisture as an **input**. Expected result: Reject it for this **prediction**-time contract.

Reason: Tomorrow's measurement is unavailable when today's **prediction** must be issued.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. next_week_wilt supplies the **training label**, not an **input** available today. The arrows show which recorded **input** or intermediate result each decision uses.

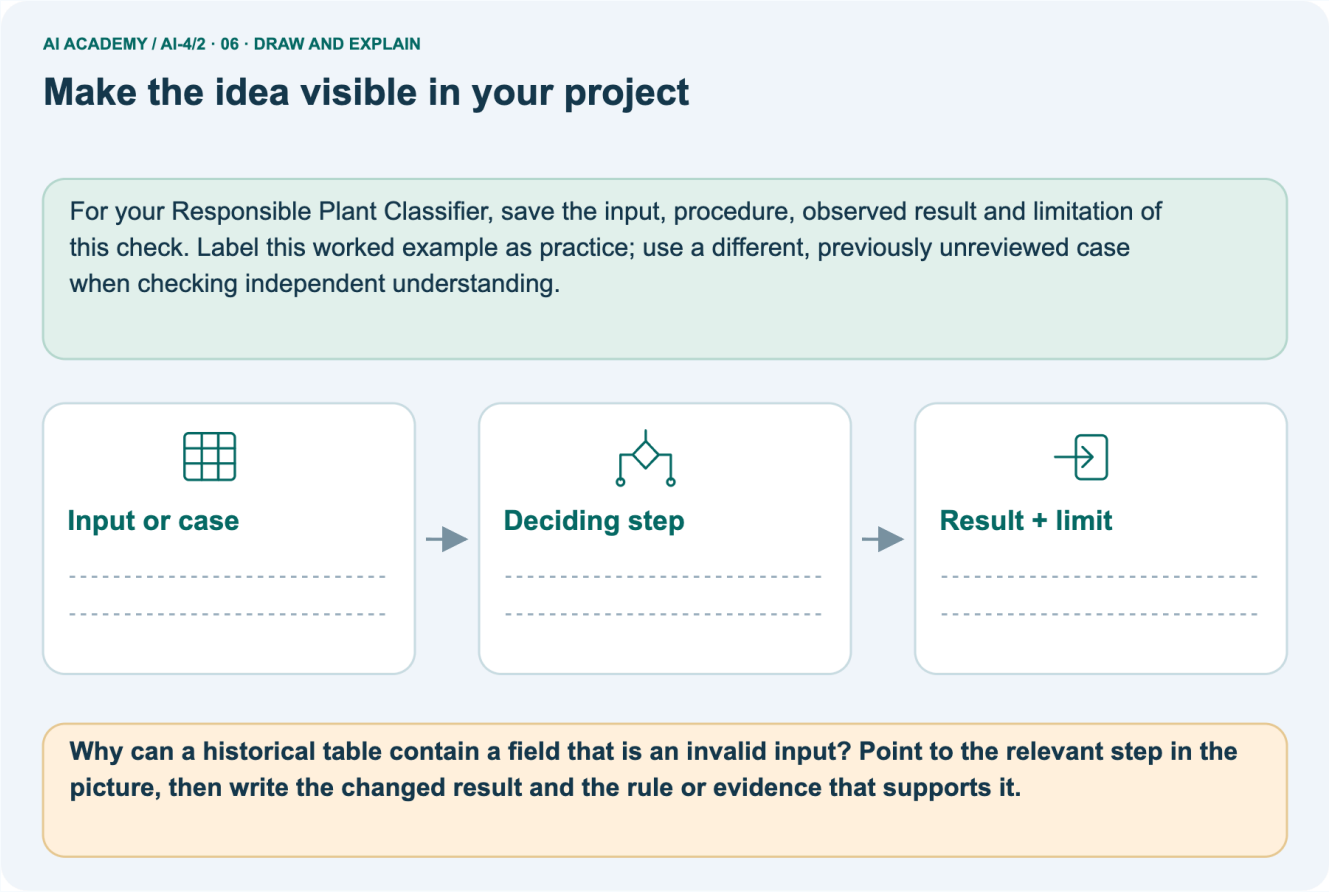
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Eligible fields are not automatically useful or causal.

Measure their predictive value with a valid split.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can a historical table contain a field that is an invalid **input**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Features, Labels, Targets: the case

02. Follow the deciding steps

03. Follow information through time

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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